

Simran Chouhan

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LinkedIn | GitHub | Portfolio

Computer Science undergraduate (AI & ML) with hands-on experience building AI-powered applications using machine learning pipelines, NLP, and retrieval-augmented generation. Comfortable with Python, Scikit-learn, and model evaluation workflows. Looking to apply practical ML skills in a real-world engineering role.

EDUCATION

- **Bachelor of Engineering in Computer Science & Engineering (AI & ML)** 2024 – 2028 (Expected)
Chandigarh University, Punjab GPA: 7.8/10.0
 - **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Artificial Intelligence, Statistics for Data Science, Database Management Systems, Object-Oriented Programming
 - **Certifications:** IBM AI Engineering (Coursera) – covers Scikit-learn, SVM, KNN, Neural Networks; IBM Full-Stack Development

PROJECTS

- **Spam Detection – Supervised Text Classification** 2025
Technologies: Python, Scikit-learn, Pandas, NumPy, NLTK, Matplotlib, Seaborn
 - Built a spam classifier on a labeled SMS/email dataset using Naive Bayes, Logistic Regression, and SVM.
 - Applied NLP preprocessing steps including tokenization, stopword removal, and TF-IDF vectorization using NLTK.
 - Compared multiple classification models and selected the best-performing one with up to 91% accuracy on the test set.
 - Evaluated models using precision, recall, F1-score, and confusion matrix to handle class imbalance in spam vs. ham data.
 - Visualized feature importance and model performance using Matplotlib and Seaborn.
- **Customer Segmentation – Unsupervised Learning** 2025
Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
 - Applied DBSCAN and Hierarchical Clustering to group unlabeled data points into meaningful clusters without predefined labels.
 - Preprocessed and scaled features using StandardScaler before applying clustering algorithms.
 - Used dendrogram analysis to determine the optimal number of clusters for hierarchical clustering.
 - Visualized cluster separation and density distributions using Seaborn and Matplotlib.
 - Compared DBSCAN's noise-handling ability against hierarchical clustering for irregular-shaped cluster detection.
- **Multilingual Voice-Enabled AI Assistant for Web Navigation** *SIH / Tekathon*
Technologies: Python, FastAPI, React, FAISS, Gemini API, LangChain, Speech-to-Text, Text-to-Speech, Docker
 - Built a RAG pipeline using FAISS vector search and Gemini API to retrieve context-aware answers from 500+ institutional documents.
 - Designed an intent classification system to map user voice queries to website navigation actions with 90%+ intent match accuracy.
 - Enabled multilingual voice interaction using Speech-to-Text and Text-to-Speech with automatic language detection across 5 languages.
 - Built a multi-level fallback system (RAG → web search → human support) reducing failed queries by over 40%.
 - Deployed FastAPI backend with React frontend and PWA support for cross-platform access.
- **TwinTalk AI – AI Mock Interview Platform** [\[Link\]](#) *Ongoing*
Technologies: React, FastAPI, MongoDB, OpenAI/Gemini API, MediaPipe, Python, Speech Recognition
 - Built a full-stack AI mock interview platform where users upload resumes or documents and the system auto-generates interview questions across Easy, Medium, and Hard difficulty levels using LLMs.
 - Implemented NLP-based text extraction from PDFs, DOCX, images (OCR via pytesseract), and screenshots to support flexible document input.
 - Developed an AI answer evaluation pipeline using OpenAI/Gemini APIs to score responses on technical accuracy, communication clarity, confidence, and completeness.
 - Integrated real-time speech recognition for voice-based interview sessions and MediaPipe for webcam-based facial expression and posture analysis.
 - Designed a post-interview report generation system producing downloadable performance reports with behavioral insights, strengths, weaknesses, and improvement suggestions.

EXPERIENCE

- Core Team Member

CodeChef Community

2024 – Present

Chandigarh University

– Contributed to UI/UX design and website management for a platform serving 500+ active students.

– Designed 20+ visual assets and event branding materials for coding contests and workshops.

– Handled website updates and supported platform stability during events with 100+ concurrent users.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript

ML / Data: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLTK, FAISS

ML Concepts: Supervised Learning, Classification, Regression, NLP, Model Evaluation, RAG, Feature Engineering

Frontend: HTML, CSS, React

Backend: Python (FastAPI), Node.js (Express)

Database: MongoDB

Tools: Git, GitHub, Jupyter Notebook, Figma, Docker